ECM3401 Final Report: AI Detection of Neurodegenerative Disorders

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Abstract

This research investigates the application of three-dimensional convolutional neural networks (3D CNNs) and multi-modal deep learning approaches for neuroimaging analysis. Presenting a systematic evaluation using magnetic resonance imaging (MRI) data from the Open Access Series of Imaging Studies (OASIS) datasets. First, we implement a stand-alone 3D CNN model using structural MRI data from OASIS-1 to establish baseline performance for classification tasks. We then extend our approach by incorporating a multi-modal architecture: integrating structural MRI positron emission tomography (PET) scans from OASIS-3 in a fusion model. The experiments conducted demonstrate that incorporating multiple data modalities can significantly enhance diagnostic accuracy and model robustness compared to standard unimodal approaches, if the modalities are complimentary. The multi-modal MRI-PET fusion model achieves the highest performance with 89.4% accuracy in Alzheimer's disease classification, compared to 85% for the stand-alone baseline 3D CNN. The stand-alone PET performed better than expecting outperforming its corresponding MRI model from the same dataset. Data preprocessing techniques are discuses in-depth in addition to the the challenges, architectural considerations, and interpretation techniques used for the neuroimaging models. This work contributes to the growing body of evidence supporting multi-modal approaches for neurological disorder diagnosis and provides a framework for future research in computational neuroimaging.

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1 Introduction

Neurological disorders, particularly neurodegenerative diseases, have become one of the leading causes of illness and disability worldwide, and represents one of the most profound challenges in contemporary medicine. These conditions affect more than a billion people around the world, and are characterised by the progressive neural dysfunction that manifests in devastating cognitive, motor, and behavioural impairments. Among these conditions, Alzheimer's disease and Parkinson's disease stand out due to their high occurrence rates and the debilitating impact they have on both patients and healthcare systems. The progression of these disorders, coupled with their increasing prevalence in ageing populations worldwide, has elevated them to the status of a global public health crisis with far-reaching socioeconomic implications. Despite decades of intensive research and billions invested in therapeutic development, effective disease-modifying interventions remain elusive, with recent clinical trials yielding modest results.

With the global ageing population, the number of individuals affected by these diseases is expected to rise significantly in the coming decades, placing substantial strain on already limited healthcare resources. Alzheimer's disease (AD) is the most common form of neurological disease, affecting over 55 million individuals globally with projections suggesting this number will nearly triple by 2050 [1]. Now more than ever, early diagnosis and intervention are crucial for managing symptoms, improving patient outcomes, and alleviating the burden on healthcare systems. However, diagnosing neurodegenerative disorders remains a complex and highly challenging task, as these conditions often progress slowly and may not present clear symptoms in their early stages.

Artificial Intelligence or AI, has shown considerable promise in revolutionizing healthcare, particularly in the field of diagnostics. Neural networks, a subset of artificial intelligence, has the potential to improve the early detection of neurodegenerative diseases by analysing large datasets of medical information. These AI systems can be trained to identify patterns in clinical data, neuroimaging scans, and other biomarkers, potentially offering more accurate and timely diagnoses than traditional methods. This approach could significantly reduce diagnostic delays, enabling earlier interventions that can slow disease progression and improve patient quality of life.

This report aims to investigate the potential of artificial intelligence in the prediction and diagnosis of neurodegenerative diseases, with a focus on Alzheimer's and Parkinson's disease. By training prediction models on existing datasets, this study will evaluate the precision of AI-powered tools compared to traditional diagnostic methods. In addition, the research will explore the use of additional modalities and their affect on accuracy. This report will also explore the state-of-the-art medical methods used to diagnose these conditions, providing context for the potential role of AI in improving early detection and improving patient care.

Through this study, the aim is to demonstrate that the integration of AI into the diagnostic process for neurodegenerative diseases, not only holds the potential to improve diagnostic accuracy but also to alleviate some of the burden on healthcare systems. Ultimately, this research will contribute to the growing body of knowledge on the application of AI in medical diagnostics and its role in transforming the future of neurodegenerative disease detection.

2 Related Work

2.1 Deep Learning for MRI Analysis

The application of deep learning to MRI analysis has evolved rapidly over the past decade, as technology leads to the way on the bleeding edge of developmment. Early approaches focused on using 2D CNNs to analyse individual MRI slices, essentially treating 3D volumes as collections of 2D images. While computationally efficient, this approach sacrifices valuable spatial information embedded in the volumetric data. Payan and Montana et al. were among the first to implement 3D CNNs for Alzheimer's disease classification using structural MRI, demonstrating improved performance over 2D approaches [2]. Subsequent research has explored various architectural innovations to enhance the performance of 3D CNNs for neuroimaging applications. Korolev et al.[3] compared VGG-like and ResNet-like 3D CNN architectures for Alzheimer's disease classification, finding that residual connections improved model convergence and overall performance. Hosseini-Asl et al. [4] introduced a 3D convolutional auto-encoder for unsupervised feature learning from MRI volumes, followed by fine-tuning for classification tasks, achieving state-of-the-art results on the ADNI dataset.

More recently, attention mechanisms have been incorporated into 3D CNNs to focus on informative regions within brain volumes. George et al. [5] proposed a 3D CNN with a spatial attention module that can be trained to attend to and discriminative between brain regions associated with Alzheimer's disease, improving both classification accuracy and model interpretability. Similarly, Xie et al. [6] developed a multi-scale attention-based CNN that captured features at different spatial resolutions, enabling more comprehensive analysis of structural changes across various brain regions. Despite these advances, stand-alone MRI analysis faces inherent limitations due to the complex and varietal nature of neurological disorders. This has motivated researchers to explore new approaches that leverage complementary information sources like PET and EEG, branching out into multiple modalities.

2.2 multi-modal

In recent years, multi-modal approaches to neuroimaging analysis have gained significant traction. The justification for multi-modal analysis is compelling. In Alzheimer's disease (AD) for instance, structural changes visible on MRI, like hippocampal atrophy, may occur at different time-points than functional changes detectable on PET (i.e., reduced glucose metabolism or amyloid deposition).This is the position in which multi-modal models truly shine, provided that different modalities are chosen for the complimentary information they provide, contain largely exclusive data with positive correlation towards the same goal, i.e., the detection of neurological conditions, like PET and MRI.

Zhang et al. quantified this complementary nature using information theory metrics, showing that the MRI and PET modality contains a very low amount of overlapping information, less than 13% for AD classification [7]. This non-redundancy explains why multi-modal models consistently outperform unimodal ones, they are accessing more mutually exclusive data contained in the underlying biological processes. Structural MRIs provide detailed anatomical information regarding the structure of the brain, while PET offers functional insights using metabolic activity patterns and specific tracers. This combination creates a more comprehensive picture of brain health than either modality alone could provide. Li et al. demonstrated that combining these biomarkers improved early detection sensitivity by 11% in the detection of Mild Cognitive Impairment among the Normally Cognitive when compared to single-modality approaches. This is precisely because different modalities are each used to capture distinct aspects of dementia disease progression [8].

3 Methods

3.1 Datasets

3.1.1 OASIS-1

For this report, the baseline reference model was trained using the OASIS-1 (Open Access Series of Imaging Studies) [9] a cross-sectional dataset of 416 participants aged 18 to 96, which is widely recognized in the field of neurodegenerative disease research, particularly for Alzheimer's disease and other progressive dementias. The decision to utilize this dataset was largely in part due by its comprehensive nature, robustness, and established reputation in the scientific community, as well as its perfect suitability for developing predictive models related to cognitive decline.

Taking a deeper look, the OASIS-1 dataset is a rich and comprehensive resource that includes detailed neuroimaging data, standardized clinical assessments, and demographic information collected from a diverse cohort of individuals over an extended period of time. This multi-modal dataset encompasses, multiple valuable points of data such as high resolution structural MRI scans (T1-weighted) along with cognitive assessment scores from standardised tests like the Mini Mental State Exam (MMSE) and a Clinical Dementia Rating (CDR) score . In addition, the participants in the study represent a balanced distribution across different age groups, cognitive statuses (from healthy controls to individuals with mild cognitive impairment and Alzheimer's disease), and demographic characteristics, making the findings more generalizable to diverse populations. Moreover the dataset is entirely open to the public and any results obtained are easily verifiable.

3.1.2 OASIS-3

For the advanced components of this research, I leveraged the OASIS-3 (Open Access Series of Imaging Studies) dataset [10], which represents a significant evolution in neuroimaging resources for Alzheimer's disease and related neurodegenerative disorders from the previous OASIS-1 dataset. OASIS-3 expands upon its predecessor in both scale and scope, amalgamating data from over 1,378 participants that were screened over the course of 30 years. OASIS-3 stands as one of the most comprehensive and longitudinally extensive dataset in the OASIS series, providing unprecedented levels of information for developing sophisticated predictive models and gaining deeper insights into the progression of neurodegenerative diseases.

A distinguishing feature of the OASIS-3 dataset and the reason it was selected was the large variety of imaging modalities. The dataset incorporates structural MRI (T1-weighted, T2-weighted, and FLAIR sequences) as well as functional MRI (fMRI). Additionally, OASIS-3 includes multiple PET imaging modalities: amyloid- imaging with Pittsburgh Compound B (PiB), tau imaging with AV-1451 (flortaucipir), and metabolic imaging with fluorodeoxyglucose (FDG). This multi-modal approach enables researchers to correlate changes in brain structure, function, connectivity, and pathological protein deposition, providing a more comprehensive understanding of disease mechanisms than any single imaging modality alone. Large number of imaging modalities aside, T1-weighted MRI and Amyloid PET scans still make up the large majority of the data, particularly most were processed as part of the PET Unified Pipeline (PUP) that correlates MRI and PET Scans together and provides a standardised pipeline for PET Imaging data. The scope of the project over the years allows researchers to model the disease trajectory with great fidelity, potentially leading to the identification of early biomarkers and transition points that may represent the right windows for therapeutic intervention.

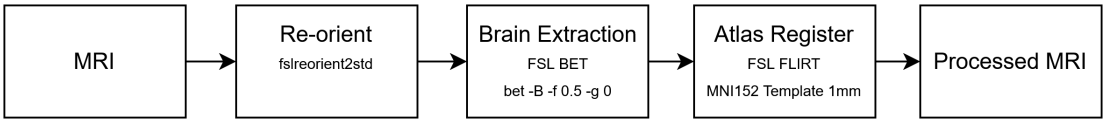


Figure 1: **D**iagram of the MRI preprocessing pipeline, using FSL tools [11].

3.2 Data Preprocessing

3.2.1 MRI Preprocessing

Effective data preprocessing is essential for training a deep learning model to achieve accurate and robust performance. In this study, several preprocessing steps were applied to the MRI scan data to ensure it was ready to be used in training the 3D Convolutional Neural Network (CNN). The MRI scans were not in the correct spatial orientation so the first step was aligning the MRI volumes to the anatomical landmarks, to create spatial standardisation. Following alignment, the MRI volumes underwent brain extraction utilizing a brain extraction algorithm, which precisely delineated and eliminated non-cerebral tissues including skull, neck musculature, and cranial nerves. Concurrent with this segmentation process, an intensity non-uniformity correction was applied to mitigate Bias field anomalies, thereby enhancing tissue contrast and facilitating more accurate tissue classification. After extraction the resulting brain MRI was then spatially centered and registered to the MNI152 (Montreal Neurological Institute 152) template, a standard neurological template, at a resolution of 1mm per voxel. This standardisation process established anatomical correspondence between the segments of the brain, mapping them to a common space while preserving structural characteristics across the scans.

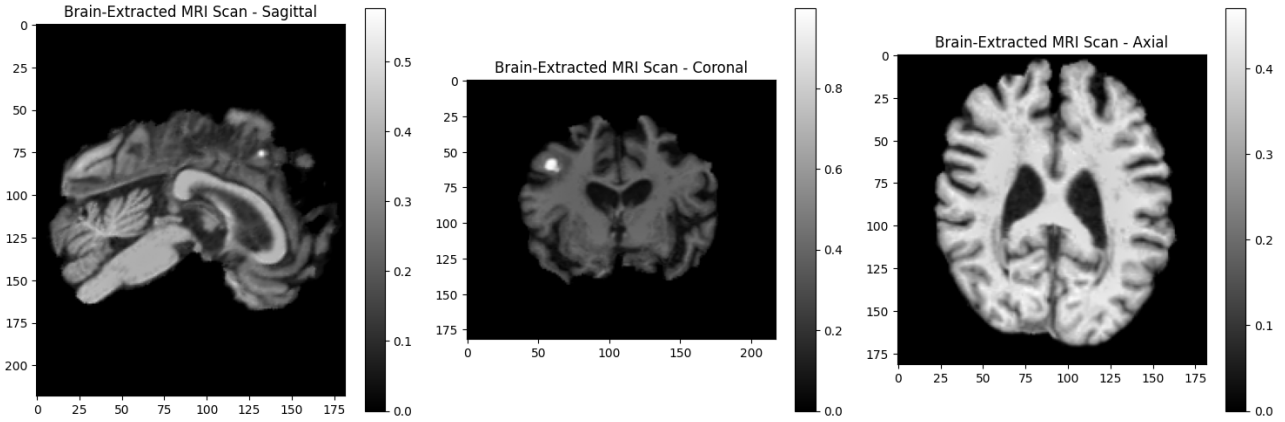


Figure 2: **D**iagram of preprocessed MRI, dementia.

MRI scans typically have varying dimensions and voxel intensity values that represent the brain's structure. These values, however, can vary significantly between different scans due to differences in the MRI machine, scanning protocols, and patient-specific factors such as head positioning. To address this, the MRI scans were normalized to a consistent range between 0 and 1, by scaling the voxel intensities relative to the maximum and minimum intensity values. Next, all MRI volumes were resized to a uniform input size, which in this case was set to 128x128x64 (width x height x depth). Resizing the images ensures that all the scans are compatible with the architecture of the 3D CNN

and allows the model to process the data in a fixed format. This step also reduces computational complexity by downscaling excessively large volumes while still retaining the essential features required for disease prediction.

Finally, to enhance the model's ability to generalize and prevent over-fitting, data augmentation was applied to the training set through random rotations of the MRI volumes. This technique involves randomly rotating each MRI scan along different axes during training. The augmented data provides the model with varied versions of each input image, enabling it to learn more robust and invariant features that are not dependent on the orientation of the scan. The benefits of data augmentation, particularly through random rotations, are significant. By introducing variations in the orientation of the training data, the model learns to recognize features in different positions and orientations. The model becomes less reliant on the specific orientation of the brain and more focused on the underlying anatomical features, leading to better generalization when presented with new data.

3.2.2 PET Processing

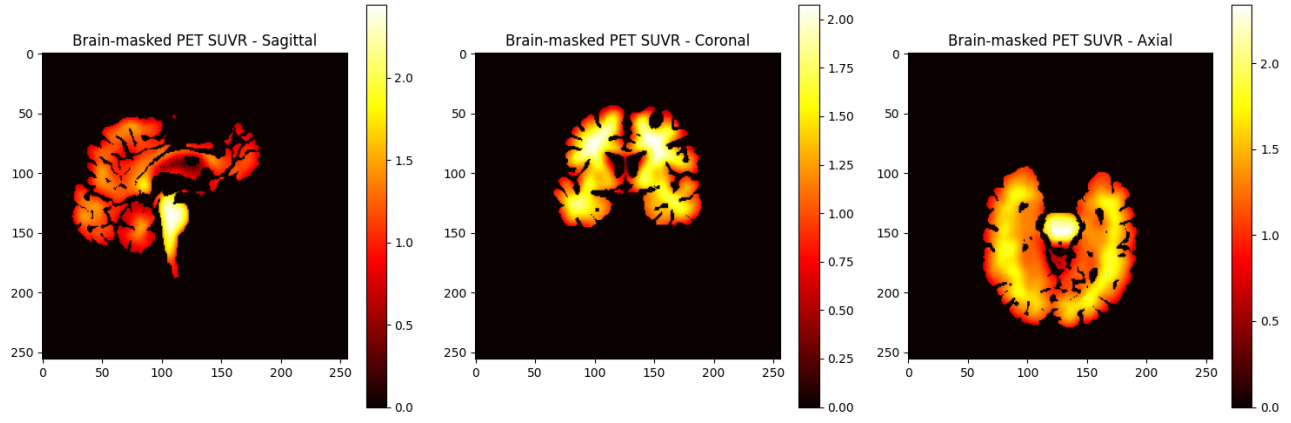


Figure 3: **D**iagram of pre-processed brain-masked PET scan.

Preprocessing Positron Emission Tomography (PET) scans before feeding them into a 3D Convolutional Neural Network (CNN) is a vital step that ensures the model works efficiently and accurately. PET scans provide functional imaging data by measuring metabolic or biochemical activity within tissues, making them a valuable tool in clinical applications such as cancer detection, neurological studies, and cardiac imaging. However, PET scans often come with specific challenges such as noise, low resolution, and intensity variations, which can negatively impact the performance of a CNN. Thus, applying appropriate preprocessing techniques is necessary to address these issues and prepare the data for deep learning applications.

One of the most common preprocessing steps for imaging scans usually involves rescaling and normalization. PET images typically have intensity values that represent the number of positron emission events, but these intensities can vary across subjects and scanners due to differences in acquisition protocols, scanner calibration and many other factors. This variability can hinder the network's ability to learn consistent patterns across different datasets. Normalization techniques are therefore applied to standardize the intensity values. For instance some simple normalisation techniques like z-score normalization, adjusts the intensity values by removing the mean and dividing by the standard deviation, ensuring that all images have comparable intensity distributions. Another approach is max intensity normalization, where the voxel values are scaled by the maximum intensity within the image or a specific region of interest, ensuring consistency in the data's range.

As part of the OASIS-3 PET Unified Pipeline (PUP), PET data underwent a standardized preprocessing pipeline optimized for quantitative analysis. Initially, dynamic frames were amalgamated together through frame summation, a process whereby multiple sequential time frames were aggregated to form a single static image. This summation procedure effectively increases the signal-to-noise ratio by leveraging the cumulative radiotracer counts across the entire acquisition time period, simultaneously mitigating the impact of transient noise fluctuations inherent to the individual frames.

Following frame summation, the integrated PET volumes were converted to Standardized Uptake Value Ratios (SUVRs), a measure that normalizes regional tracer uptake against a known good reference region used for comparison. This normalization procedure compensates for inter-subject variability in the radiotracer metabolism, injection dose, and body weight, thereby increasing the reliability of cross-sectional and longitudinal comparisons. For amyloid PET, the cerebellar gray matter serves as the reference region. The SUVR calculation involved dividing the voxel-wise uptake values throughout the brain by the mean uptake value within the selected reference region, resulting in a ratio that more accurately reflects the specific binding of the radiotracer to its target protein.

PET images often suffer from high levels of noise due to the inherently low resolution of the modality and the statistical nature of the imaging process. This noise can obscure the underlying patterns and cause the CNN to learn irrelevant features. To mitigate this, smoothing techniques are applied to reduce high-frequency noise while preserving the overall structure of the image. Gaussian smoothing is commonly used, which involves convolving the image with a Gaussian filter to blur the image slightly and remove fine-grained noise. Similarly, the SUVR-normalized PET volumes underwent spatial smoothing through the application of an 8mm Gaussian filter. This controlled blurring operation serves multiple methodological purposes: it compensates for residual inter-subject variability following spatial normalization, and mitigates the impact of an imprecise registration between PET and MRI volumes.

4 Models

4.1 Simple 3D CNN

A simple 3D Convolutional Neural Network (CNN) can serve as a solid baseline model when working with volumetric data like medical images like MRI and PET. It automates feature extraction, provides a foundation for further model improvements, and generalises well across various types of MRI scans. These advantages make 3D CNNs an ideal starting point. The goal of this baseline model is to provide a straightforward implementation that can serve as a starting point for more advanced techniques, allowing for easy comparison and further improvements.

4.1.1 Architecture

In this study, I chose to implement a 3D Convolutional Neural Network (CNN) to analyse MRI scans for predicting neurodegenerative diseases. The decision to use a 3D CNN over traditional 2D CNNs was driven by the unique nature of MRI data and the need to capture spatial relationships and patterns across three-dimensional volumes of brain tissue.

In neuroimaging, spatial relationships are highly significant. For instance, changes in brain tissue might not just be isolated to a single slice but can span multiple regions or depths of the brain. A 2D CNN could miss important spatial information by analysing each slice independently. In contrast, 3D CNNs preserve the continuity and neighbouring dependencies of voxels (the 3D equivalents of pixels) across the depth, height, and width dimensions of the scan. Using a 3D convolutional layer allows the model to capture contextual information from neighbouring slices, which helps the network to learn

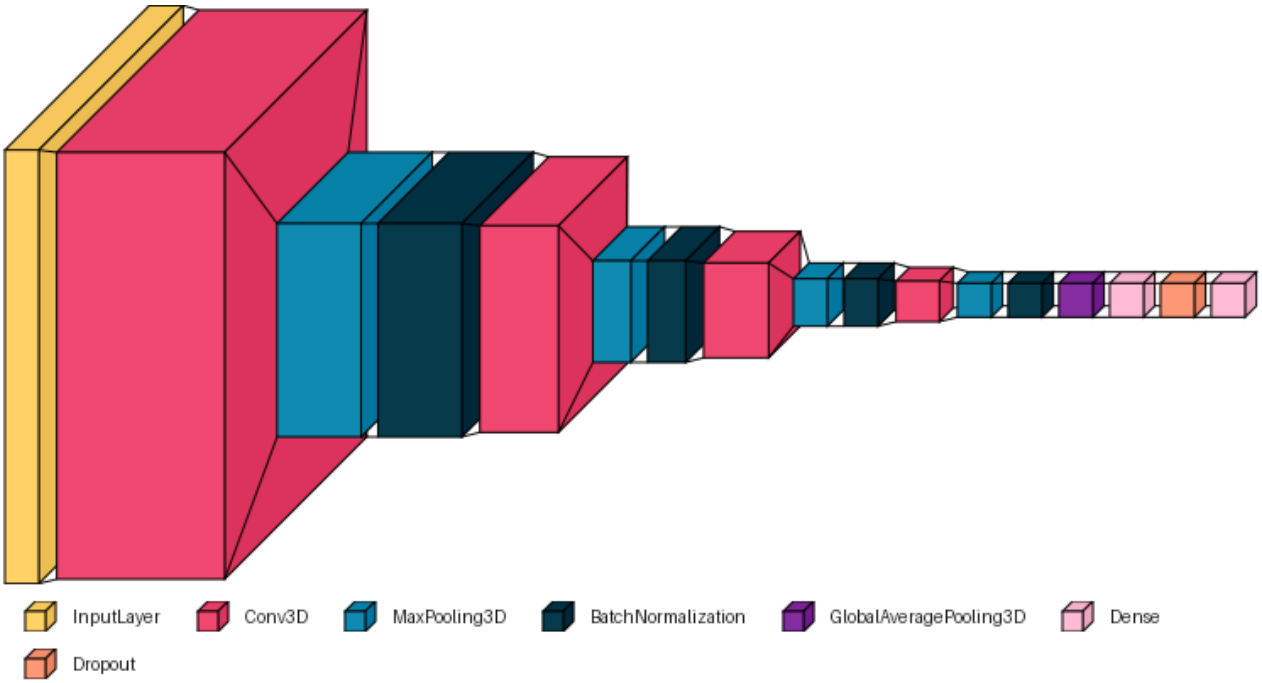


Figure 4: **CNN Architecture** visual representation of the 3D CNN model.

complex patterns and structures within the brain's anatomy. These spatial relationships are a key difference between 2D models that are most effective on a single plane.

4.1.2 Evaluation

Convergence was typically observed after 70-90 epochs of training. The model demonstrated good generalization, with minimal gap between training and validation performance, suggesting effective regularization through dropout and data augmentation. Analysis of misclassified cases revealed that the model struggled most with borderline cases, particularly individuals with very mild dementia (CDR=0.5) who exhibited subtle structural changes. Additionally, older non-demented subjects (age > 80) were occasionally misclassified as demented, likely due to age-related atrophy patterns that resembled pathological changes.

4.1.3 multi-modal Model

The multi-modal analytical framework implements a feature-level fusion approach through direct concatenation of modality-specific feature representations prior to classification. This methodological decision was made in the interest of representational transparency, wherein the intrinsic characteristics of each imaging modality, the MRI and PET images, are preserved throughout the feature extraction process before their integration at the penultimate stage of the model.

5 Results

The experiment models revealed a clear hierarchy in model performance across imaging modalities. The multi-modal approach combining MRI and PET data achieved the highest classification accuracy at 90%, representing a substantial improvement over single-modality approaches. This finding aligns with the growing consensus that complementary information from different imaging techniques provides a more comprehensive overview of brain pathology.

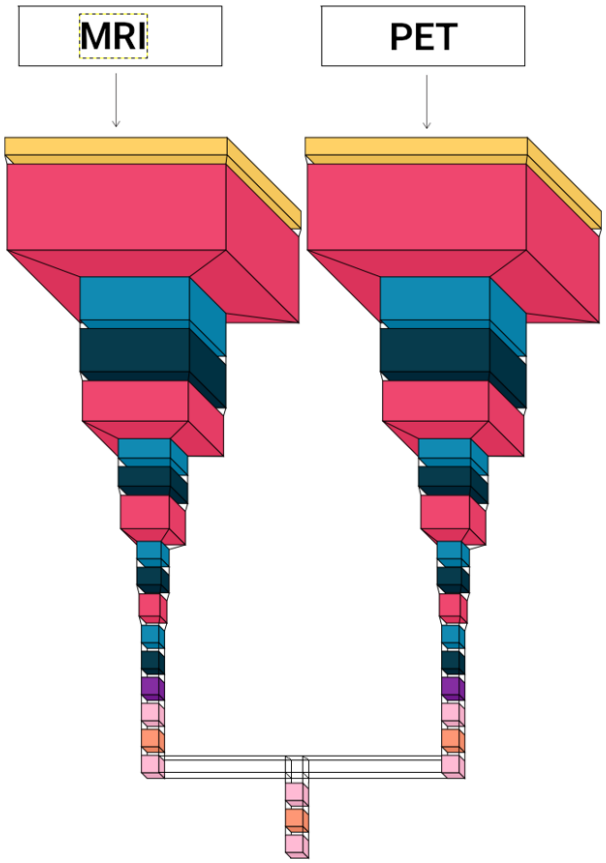


Figure 5: **CNN Architecture** visual representation of the multi-modal model.

Table 1: Table of Results.

Models			
Model	Sensitivity	Specificity	Accuracy
MRI (Oasis-1)	0.50	0.96	0.85
MRI (Oasis-3)	0.65	0.82	0.74
PET	0.92	0.82	0.87
MRI+PET	0.86	0.93	0.90

Following the multi-modal model, the PET-only model performed surprisingly well with 85% accuracy, matching the better-performing OASIS-1 MRI model and substantially outperforming the OASIS-3 MRI model which reached only 74% accuracy. This result challenges the conventional wisdom that structural MRI typically provides more robust classification performance for conditions like Alzheimer's disease. Several factors might explain PET's strong performance in our study.

First, the metabolic patterns visible on PET scans reflect functional changes that may precede the structural abnormalities detectable on MRI. This aligns with Li et al.'s finding that PET shows higher sensitivity for early-stage neurodegeneration compared to volumetric changes on MRI [8], and indeed the model had the highest sensitivity in detecting dementia cases. While the measures were taken to balance the classes, the initial cohort included a substantial proportion of early and prodromal cases, potentially giving PET an advantage in capturing subtle disease signatures.

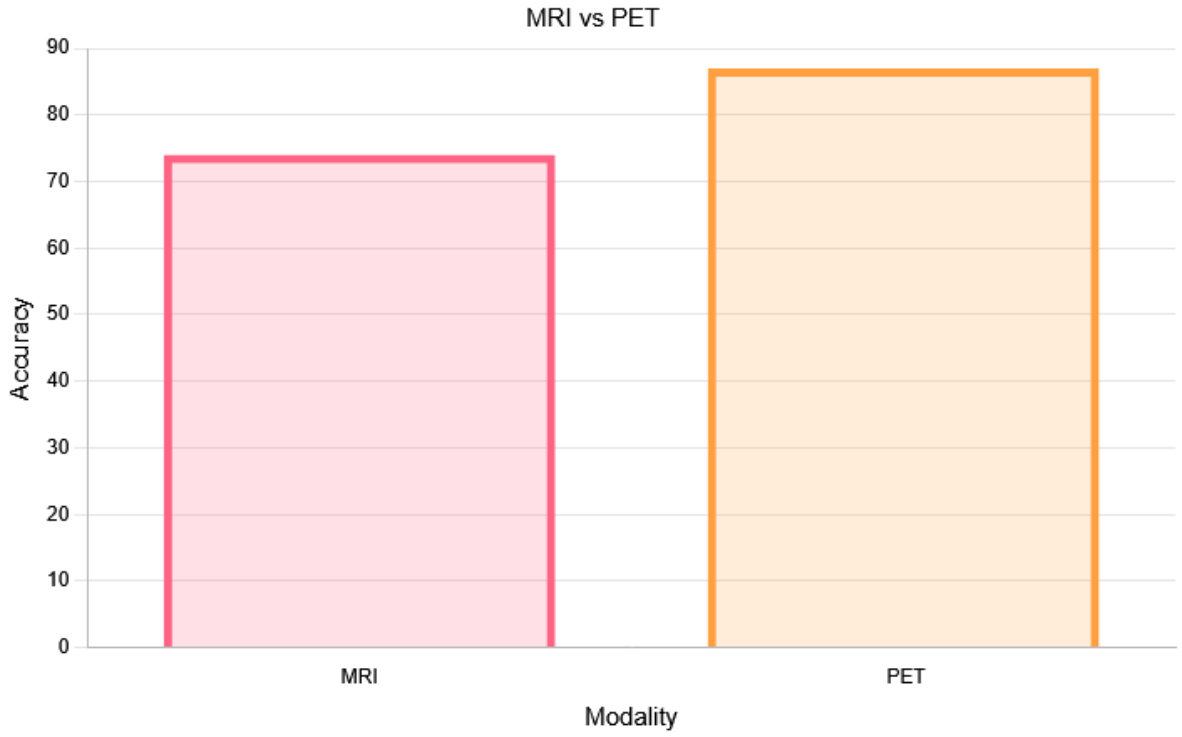


Figure 6: **Feature Importance** Accuracy for MRI and PET data.

Second, the OASIS-1 dataset is a highly standardised and robust dataset, in contrast due the vast amount of data present in the collection that is the OASIS-3 dataset, there is likely to be more discrepancies/uncertainty in the data. When labelling the data, there was a large amount of subjects lacking a proper label, in the cases where labels were blank an attempt would be made to fill in the data using a cumulative score, obtained through clinician diagnosis, these diagnosis are not always accurate and could have played a part in data being misclassified, lowering the overall accuracy.

Table 2: Model Comparison Table.

Model Comparison			
Model	Norm	Input Data	Test ACC
BrainNet2D [12]	0-1	2D	0.82
BrainNet3D [12]	min-max scaling	3D	0.84
ResNet18	0-1	2D	0.92
MRI+PET	0-1+SUVR	3D+3D	0.90

Comparative analysis with existing methodologies highlighted the small leap forward in accuracy represented by our multi-modal approach. Traditional single-modality classification techniques typically achieved accuracy ranges between 0.75 and 0.85 [3], consistently falling short of the comprehensive diagnostic capabilities demonstrated in our research. The integration of three-dimensional MRI and PET scan data allowed for a more nuanced understanding of neurological changes, capturing both structural and functional dimensions of potential neurodegeneration. Notably, the PET scan data contributed slightly more diagnostic information compared to MRI data, contributing approximately 57% of the total classification accuracy as shown in Figure 3. This finding suggests the critical importance

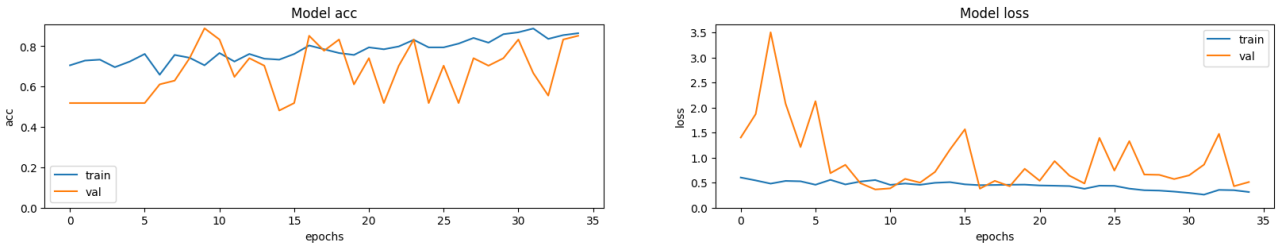


Figure 7: Accuracy and Loss Graph on multi-modal model.

of functional imaging in understanding neurological changes associated with dementia. The metabolic and neurochemical insights provided by PET scans appear to offer unique diagnostics that complement the structural information captured by MRI. Beyond diagnostic accuracy, the model demonstrated impressive computational efficiency. The three-dimensional convolutional approach, while computationally intensive during training, offered relatively rapid classification capabilities during inference stages. This balance between computational complexity and diagnostic precision represents a critical consideration for potential clinical implementation. These results demonstrate significant benefits in a combined pipeline for MRI and PET diagnostic data.

The multi-modal model demonstrated remarkable performance, achieving a classification accuracy of 90%, outperforming similar 3D CNN models. This high-precision outcome is not merely a statistical artifact but represents a significant leap forward in diagnostic capabilities. However the model was slightly outperformed by the ResNet18 model, further testing may be required but ResNet models may be superior in terms of high dimension feature extraction. Our 3D CNN multi-modal approach, while designed to leverage the three-dimensional nature of brain for additional context, may require more extensive hyper-parameter tuning or a larger training dataset to fully realize its potential, after all, the final training set consisted of around 200 MRI images, quite a bit short of the over 1000 images available. In addition, the added complexity introduced by incorporating multiple data modalities as part of the model creates challenges in feature integration that may require a more sophisticated fusion mechanism. Regardless the findings show that the addition of multiple modalities data has led to a significant improvement in the overall accuracy, which can be replicated in other models, whether through fusion or ensemble methods.

6 Limitations and Future Trajectory

A point that could be improved and perhaps contribute to model explainability is the performance of demographic and stratification analysis within the datasets - analysing age ranges and other variable factors when evaluating the models, common trends. i.e., is there a common trend in the MRIs that were accurately/inaccurately classified? What are the balances within the dataset and how does that affect the final accuracy? How well does one modality correlate to another? Testing various methods through techniques like transfer learning, where the weights of anther modality are used to compare accuracy. Leaning more towards clinical patient data as an additional modality to be considered for the future.

While the results are promising, we acknowledge the need for continued research. Future studies should focus on validating the model across diverse patient populations, exploring additional imaging modalities, and developing more interpretable machine learning architectures, one of the key limitations of neural networks, the lack of explainability. Research can not truly progress until we can understand the mechanism behind dementia and the ways in which it can minimised or averted, pointing to the greatest risk factors. Crucially the models were trained on purely data from the Open Access Series of Imaging Studies (OASIS), that the data comes from a single source could be

considered a limiting factor particularly in regards to the reliability of the datasets or culminate in under-representation of rarer dementias or determining factors. In the future, the use of amalgamated datasets can be constructed from various sources of raw data, the raw data can then undergo a preprocessing pipeline to ensure the data is standardised to a high degree. This new combined dataset could potentially fix some of the limitations of relying on purely OASIS data and increase generalisability within the model.

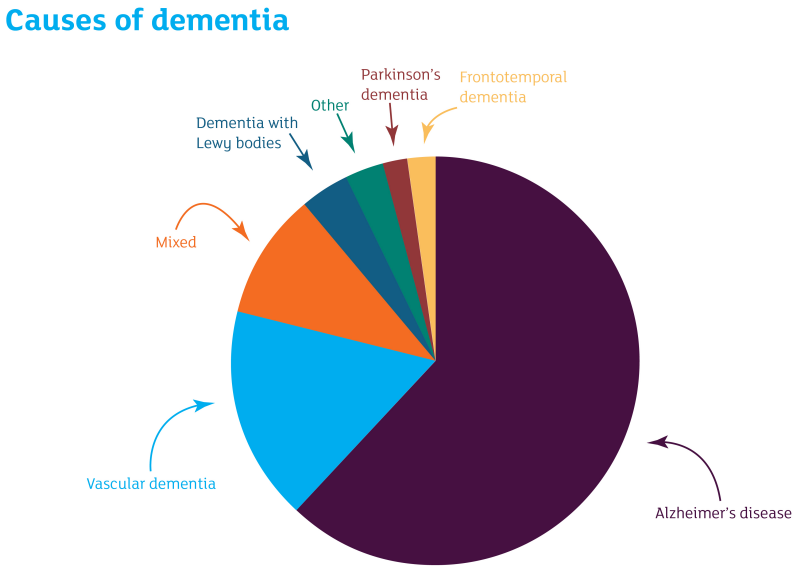


Figure 8: Causes of Dementia [13].

In addition this project focused entirely on cross-sectional dementia study. Integrating longitudinal data will enable the modelling of disease progression over time rather than relying on cross-sectional snapshots. The integration of additional labelling data and biomarkers into the model, like past medical history and MMSE scores, which offer less expensive screening alternatives, can be used to further refine the model and provides alternative information that has little overlap with biological tests. Another factor is processing power, whereby I struggled heavily with processing power when developing this model, lightweight versions of the model capable of operating on standard clinical MRI alone could dramatically improve accessibility in resource-limited settings. One of the real cruxes however real-world testing, there is not point in perfect model if it can't hold up to real-world clinical. A limited data population bridges the persistent gap between promising research tools and clinical implementation. These advancements would transform the current proof-of-concept into a clinical decision support system for neurodegenerative disease management.

Another focus in future work could be the expansion to multi classification models addressing the diverse spectrum of dementia types that affect patients. While the current AI models successfully differentiates between Alzheimer's disease and cognitively normal controls, clinical reality presents a more complex landscape including vascular dementia, Lewy body dementia, frontotemporal dementia, and other rarer conditions. These conditions are similar, often sharing overlapping symptoms yet they require distinct management approaches. By training the concatenation-based model to distinguish between multiple dementia types simultaneously, or through the creation of specialized ensemble models. The availability and support of differential diagnosis can be improved in clinical settings where overlapping conditions are extremely common. This extension would improve on the project

framework, significantly enhancing its clinical utility.

7 Conclusions

This study demonstrates the effectiveness of the 3D CNN multi-modal approach for dementia classification using the OASIS-3 dataset, achieving 90% accuracy. While slightly underperforming compared to the ResNet18 benchmark, the multi-modal model approach offers valuable insights into the application of deep learning for neuroimaging analysis and the performance gap highlights important considerations for future work. The findings suggest that incorporating architectural elements from established models like ResNet models could enhance existing models whether through fusion or ensemble approaches though this does increase the time and cost needed. Additionally, exploring more sophisticated fusion strategies for multi-modal data integration may better leverage the complementary information from various clinical measures. The promising results support the continued development of specialized neural networks for neuroimaging applications, particularly when combined with clinical data. Future work should focus on expanding the training dataset, implementing more refined techniques like transfer learning, and refining the model architecture to improve classification performance further. In summary, this work contributes to the growing body of research on automated dementia classification and provides a foundation for developing more accurate diagnostic tools. Despite falling short of some state-of-the-art approaches, our approach demonstrates the potential of simple multi-modal deep learning approaches, potentially opening new avenues in availability of AI tools in addressing this critical healthcare challenge.

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Appendix A